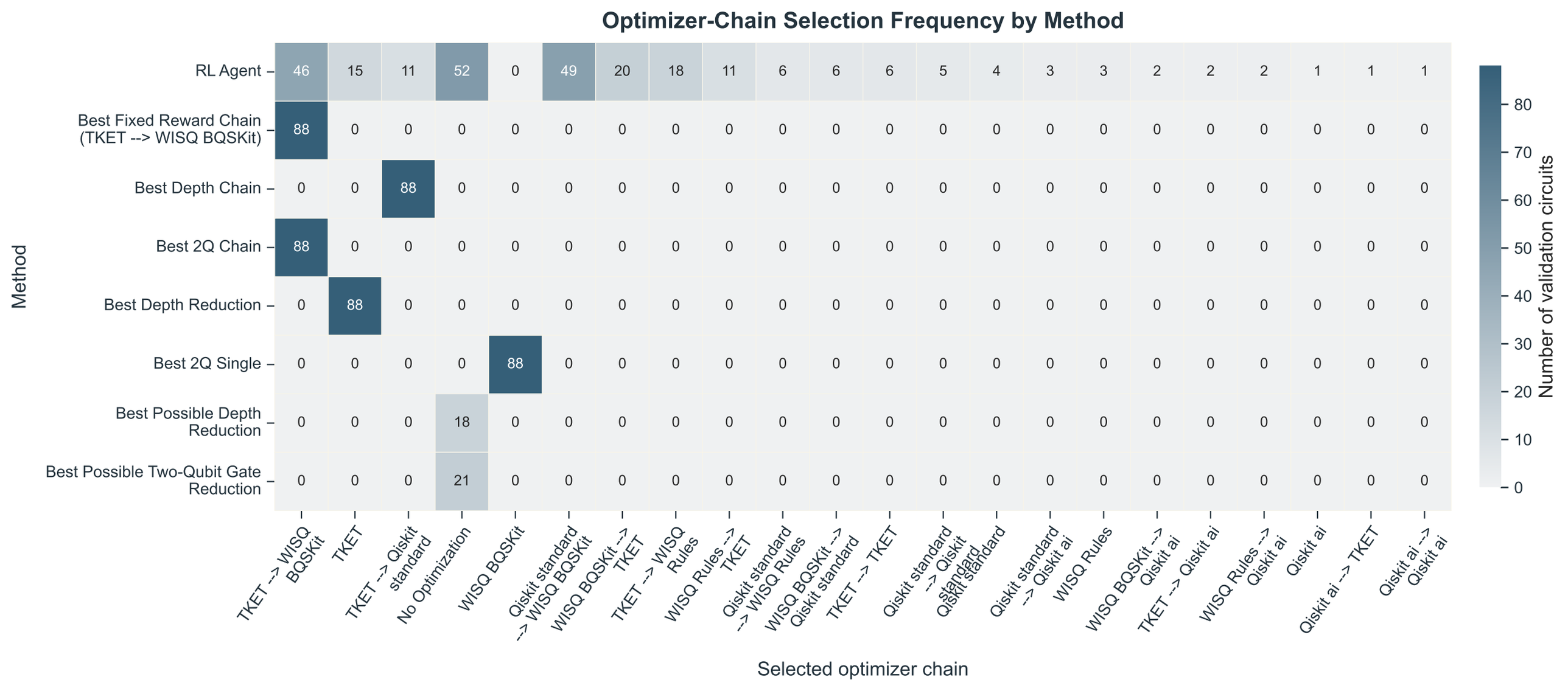
Optimizer-Chain Selection Frequency by Method
Method
RL Agent
Best Fixed Reward Chain (TKET --> WISQ BQSKit)
Best Depth Chain
Best 2Q Chain
Best Depth Reduction
Best 2Q Single
Best Possible Depth Reduction
Best Possible Two-Qubit Gate Reduction
46 15 11 52 0 49 20 18 11 6 6 6 5 4 3 3 2 2 2 1 1 1
88 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 88 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
88 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 88 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 88 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 18 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 21 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
TKET --> WISQ BQSKit
TKET
TKET --> Qiskit standard
No Optimization
WISQ BQSKit
Qiskit standard --> WISQ BQSKit
WISQ BQSKit --> TKET
TKET --> WISQ Rules
WISQ Rules --> TKET
Qiskit standard --> WISQ Rules
WISQ BQSKit --> Qiskit standard
TKET --> TKET
Qiskit standard --> Qiskit standard
Qiskit standard --> Qiskit ai
WISQ Rules
WISQ BQSKit --> Qiskit ai
TKET --> Qiskit ai
WISQ Rules --> Qiskit ai
Qiskit ai
Qiskit ai --> TKET
Qiskit ai --> Qiskit ai
Selected optimizer chain
Number of validation circuits
0
10
20
30
40
50
60
70
80